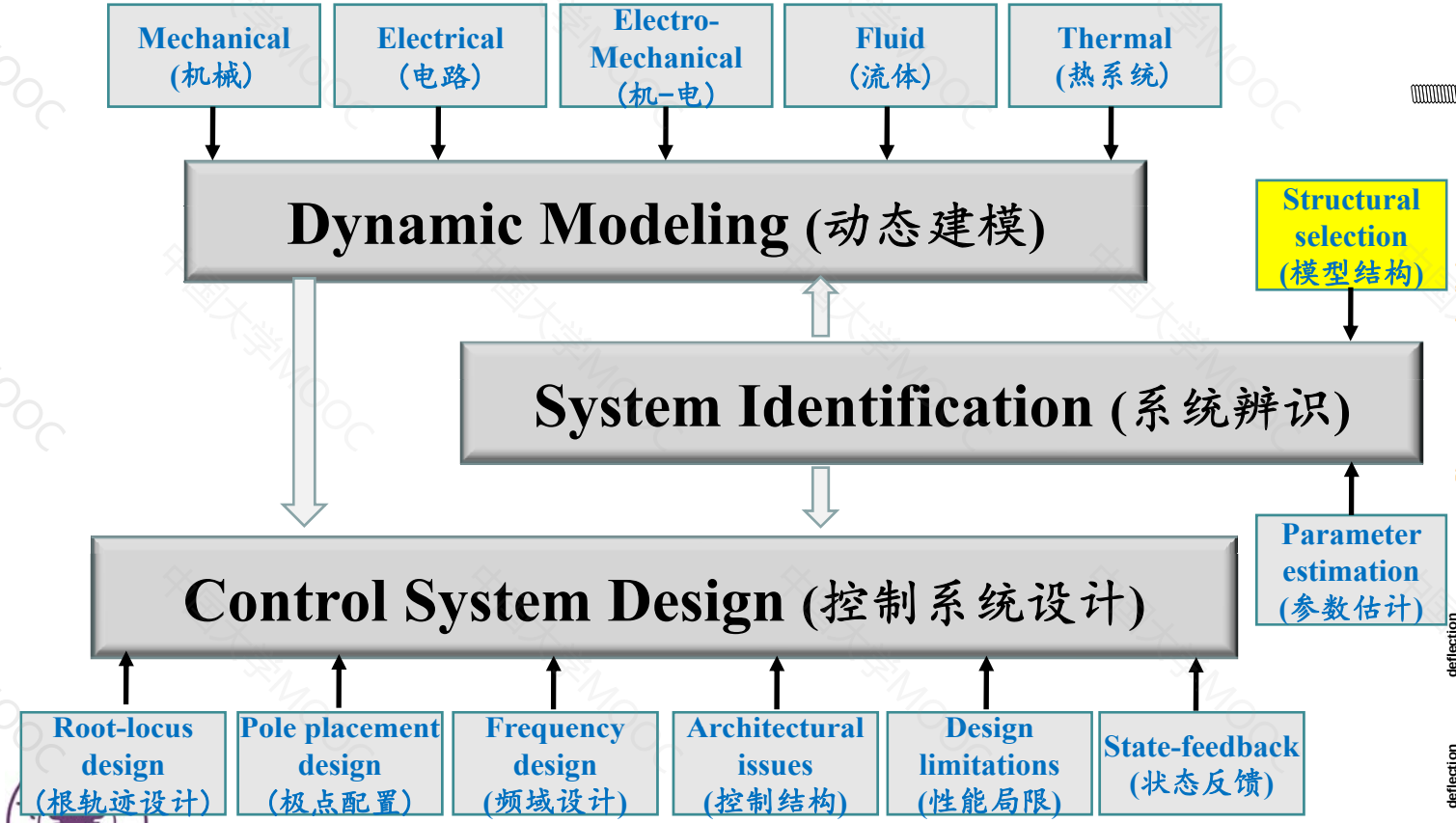
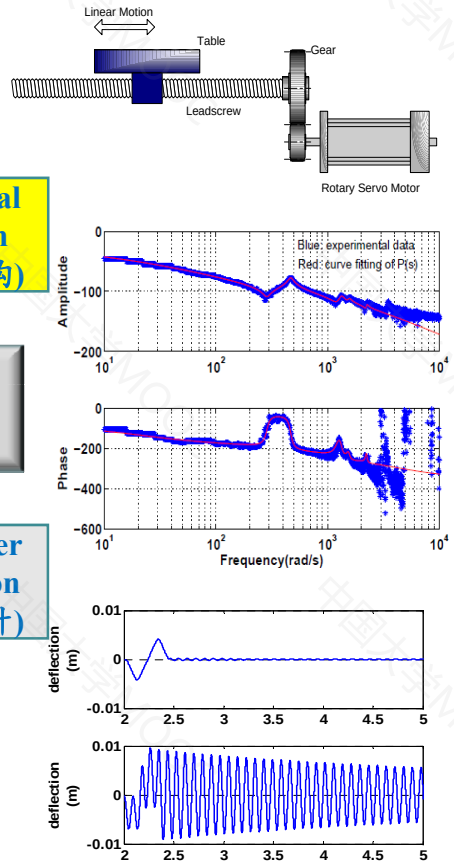
General Procedure of System Identification

(系统辨识的一般过程)

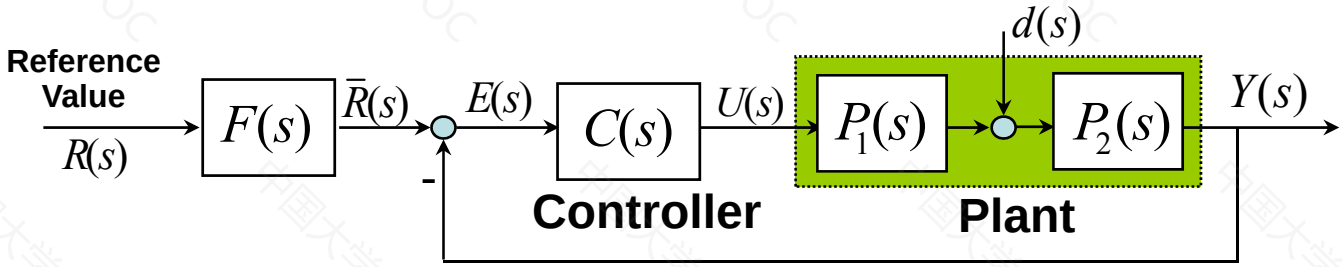
Main Contents of Course (课程主要内容)





System Identification

- **Concept of System Identification** (系统辨识的概念)



– Dealing with the problem of building mathematical models of dynamical systems ($P(s)$) based on observed responses of the system (基于观测到的系统响应建立动态系统数学模型的问题)

– First step in any model-based control design process (任何基于模型的控制设计过程的第一步)

– Quality of the resulting model has significant practical importance to the achievable closed-loop control performance (所得结果模型的品质对可获得的闭环控制性能有重要的实际意义)

General Procedure and Issues

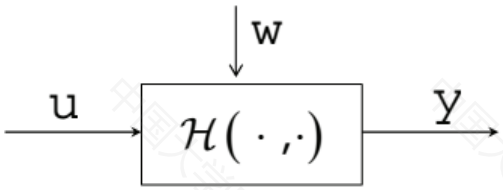
- **Dynamic System** (动态系统)

$$y = \mathrm{H}(u, w), u \in \Omega_u, w \in \Omega_w, \mathrm{H} : \ell_{\infty e} \times \ell_{\infty e} \to \ell_{\infty e}$$

y: output (输出)

u: input and/or measured disturbance (输入和/或测量到的干扰)

w: unmeasured disturbance (未测量的干扰)



- **System Identification** (系统辨识)
  - Obtaining the mapping *H* based on experimental data collected through conducting certain experiments (通过实施某些实验并基于所收集到的实验数据来获得映射***H***)
- **Implementation** (实施方法)
  - Constructing an estimation of *H* based on the collected experimental data set $Z_t = \{U_t, Y_t\}$ at sampling instant time *t*, in which $U_t = \{u(t), u(t-1), \ldots\}$ and $Y_t = \{y(t), y(t-1), \ldots\}$ (基于在采样瞬时时间***t***所收集到的实验数据$\boldsymbol{Z_t = \{U_t, Y_t\}}$来构建***H***的估计量)

ID Step 1— Model Structure Selection

- **Selection Considerations**(选择依据)
  - Compact model which can be stored and used easily (能存储和易使用的紧凑模型)
  - Adequately describe the actual system's behavior for its intended use (能恰当描述实际系统的行为)
  - Restricting the model within a set of known models or model structures which can be conveniently stored and easily analyzed (将模型限制在一系列易于存储和分析的已知模型或模型结构中)

- **Parametric Model Set or Structure** (参数化模型集或结构)

$$M := \left\{ M(\theta) : \; M(\theta) = G_\theta(u, w), \theta \in \Omega_\theta \subset R^p \right\}$$

where G_θ is a known model depending on θ, θ is a vector of finite dimension p (**G_θ是一个依赖于θ的已知模型，θ是一个有限维向量**)

Factors Influencing Model Set Selection

- **1.Ultimate Purpose of System Identification (系统辨识的最终目的)**
  - **Models for applications such as realistic simulations (模型用于真实系统仿真)**

    (i) need to be large and complex enough to describe various system behaviors precisely (模型足够大且足够复杂以便能精确描述各种系统行为)

    – ***model error*** issue (模型误差问题)

    (ii) should not be overly complex to uniquely determined model parameters based on available experimental data (模型不要过于复杂以便易于基于实验数据唯一确定模型参数)

    – ***identifiability*** issue (可辨识问题)
  - **Models for applications of model-based controller designs (模型用于控制器设计)**

    (i)capture the essential information needed by the chosen controller design methodology (抓住所选择的控制器设计方法中所需的本质信息)

    (ii) simple enough for easily use in controller design and implementation, in which on-line computation cost is a major concern (足够简单以便易于控制器设计和实施，其中在线计算量是一个主要的考虑因素)

Factors Influencing Model Set Selection

- **2.Collectable/Available Experimental Data (可收集/获得的实验数据)**

(i) The experiments that can be performed are limited (e.g. on-board modeling) (可做的实验受限，如在线建模) → Experimental data may not be so rich to uniquely determine all parameters of a generic model structure (实验数据可能不太丰富而不能确定通用模型结构中的所有参数) → The model structure are further constrained to satisfy parameter identifiability conditions, which might be at the expanse of losing some generality and accuracy of the identified models (可能会牺牲所选择模型的通用性和精确性而使模型结构满足参数可辨识性条件)

(ii) The chosen model structure is too small that it can never capture all the important characteristics (所选择的模型结构太小而不能反映系统的重要特性) → The model structure needs to be enlarged instead to reduce model error (模型结构需要增大以降低模型误差)

Common LTI Model Structures

- **Auto Regressive with Exogenous Input (ARX) Model**
  (自回归各态历经模型)

$$y(t)+a_1y(t-1)+\ldots+a_{n_a}y(t-n_a)$$
$$=b_0u(t-d)+b_1u(t-d-1)+\ldots+b_{n_b}u(t-d-n_b)$$

➢ Regressor form**(回归形式):** $y(t)=\varphi(t)^{\mathrm{T}}\theta$

where, $\varphi(t)^{\mathrm{T}}=\left[-y(t-1),\ldots,-y(t-n_a),u(t-d),\ldots,u(t-d-n_b)\right]$

$$\theta=\left[a_1,\ldots,a_{n_a},b_0,\ldots,b_{n_b}\right]^{\mathrm{T}}\in R^{n_a+n_b+1}$$

➢ Concise notations**(简明表示):**

$y(t-1)=q^{-1}y(t),\ldots\ldots,$

$$A(q^{-1})y(t)=q^{-d}B(q^{-1})u(t),\quad \text{or}\quad y(t)=G(q^{-1})u(t)$$

where

$$G(q^{-1})=\frac{q^{-d}B(q^{-1})}{A(q^{-1})},\qquad A(q^{-1})=1+a_1q^{-1}+\ldots+a_{n_a}q^{-n_a}$$
$$B(q^{-1})=b_0+b_1q^{-1}+\ldots+b_{n_b}q^{-n_b}$$

Common LTI Model Structures

- **ARMAX Model** (ARMAX模型)

$$A(q^{-1})y(t)=q^{-d}B(q^{-1})u(t)+C(q^{-1})e(t)$$

- **Output Error(OE) Model** (输出误差模型)

$$y(t)=\frac{q^{-d}B(q^{-1})}{L(q^{-1})}u(t)+e(t)$$

- **Box-Jenkins (BJ) Model** (BJ模型)

$$y(t)=\frac{q^{-d}B(q^{-1})}{L(q^{-1})}u(t)+\frac{C(q^{-1})}{D(q^{-1})}e(t)$$

- **General SISO LTI Model** (一般单输入单输出线性时不变模型)

$$A(q^{-1})y(t)=\frac{q^{-d}B(q^{-1})}{L(q^{-1})}u(t)+\frac{C(q^{-1})}{D(q^{-1})}e(t)$$

$e(t)$: a white noise sequence (白噪声)

$$C(q^{-1})=1+c_1q^{-1}+\ldots+c_{n_c}q^{-n_c}$$

$$L(q^{-1})=1+l_1q^{-1}+\ldots+l_{n_l}q^{-n_l}$$

$$D(q^{-1})=1+d_1q^{-1}+\ldots+d_{n_d}q^{-n_d}$$

ID Step 2 — Criterion of Fit

- **Natural Choice** (自然选择)
  - Minimize the discrepancy between the measured output and the predicted output of the model using the same set of past input and output measurement data (在相同的输入和输出数据下使测量输出和模型预测输出之间的差值最小).

- **Actual Implementation** (实际实施)

Even using the same set of past input and output measurement data, as the model structure may contain some random signals, the predicted output of the model will not be unique in reality (即使采用同样的输入和测量输出数据，由于模型结构可能包含一些随机信号，模型的预测输出实际上将不唯一).

To solve this ambiguity, the optimal predictor of a model structure based on the past measurement data will be used to fit the measured output of the actual system instead (为解决这个不确定性，将采用基于过去测量数据的模型结构优化预估器来拟合实际系统的测量输出).

Optimal Predictor

- ***j* Step Ahead Predictor (*j*步超前预估器)**

$$\hat{y}(t+j\,|\,t)=f[\theta,y(t),y(t-1),\ldots,u(t),u(t-1),\ldots]$$

- ***j* Step Ahead Optimal Predictor of a Model (模型的*j*步超前优化预估器)**

Minimize $$E\{[y_m(t+j)-\hat{y}(t+j\,|\,t)]^2\}$$

E: the expectation operator (数学期望)

$y_m(t+j)$: the output of the model structure at instance $t+j$ (在$t+j$时刻的模型结构输出)

- ***j* Step Ahead Optimal Predictor of ARMAX Model (ARMAX模型的*j*步超前优化预估器)**

$$A(q^{-1})y(t)=q^{-d}B(q^{-1})u(t)+C_e(q^{-1})e(t)$$

$$\hat{y}(t+j\,|\,t)=\frac{F_j(q^{-1})}{C(q^{-1})}y(t)+\frac{J(q^{-1})B(q^{-1})}{C(q^{-1})}u(t-d+j)$$

Pseudolinear regression form(伪线性回归形式):

$$\hat{y}(t+j\,|\,t)=\varphi(t,\theta)^T\theta$$

$$\theta=[f_{j0},\ldots,f_{jn_f},g_0,\ldots,g_{n_b+j-1},c_1,\ldots,c_{n_c}]^T,$$

$$G_j(q^{-1})=J(q^{-1})B(q^{-1})=g_0+g_1q^{-1}+\ldots+g_{nb+j-1}q^{-(nb+j-1)}$$

$$\varphi(t,\theta)^T=\left[y(t),\ldots,y(t-n_f),u(t-d+j),\ldots,u(t-d-n_b+1),-\hat{y}(t-1+j\,|\,t-1),\ldots,-\hat{y}(t-n_c+j\,|\,t-n_c)\right]$$

Criterion of Fit – Least Squares

- **Optimal Predictor of Chosen Model Structure** (所选择模型结构的优化预估器)

$$\hat{y}(t+j\,|\,t)=f[\theta,y(t),y(t-1),\ldots,u(t),u(t-1),\ldots]=\varphi(t,\theta)^T\theta$$

- **Criterion of Fit** (拟合准则)

$$J(t,\theta)=\sum_{i=1}^{t-j}[y(i+j)-\hat{y}(i+j\,|\,i)]^2=\sum_{i=1}^{t-j}[y(i+j)-\varphi(i,\theta)^T\theta]^2$$

- **Parameter Estimation Algorithm** (参数估计算法)
  - Solving the following optimization problem (求解以下优化问题)

$$\hat{\theta}(t)=\arg\{\min_{\theta}J(t,\theta)\}$$

  - Numerical solution via iteration in general as the performance index $J(t,\theta)$ is a nonlinear function of θ (由于性能指标$J(t,\theta)$是θ的非线性函数，通常通过迭代数值求解)
  - Local minimum solution likely (倾向于局部最小求解)

ID Step 3 — Experimental Design

- **Identifiability Requirement** (可辨识性要求)
  - The data set is informative enough to uniquely determine model parameters (要求数据集具有足够量信息以便唯一确定模型参数)

- **Minimizing Model Error Effect** (最小化模型误差影响)
  - The characteristics of the model error depends on the chosen model structure (模型误差特性依赖于所选择的模型结构)
  - Actual magnitude of the model error depends on the specific experiments performed (实际模型误差幅度依赖于特定进行的实验): use removing bad data and offsets, filtering, segmentation of data to obtain a date set that minimize the effect of the model error on the consistency and/or accuracy of the identified model parameters (去掉不良数据和偏移量、滤波、数据分段等来获得数据集用于辨识，从而使模型误差对所辨识模型参数的一致性和准确性的影响最小).

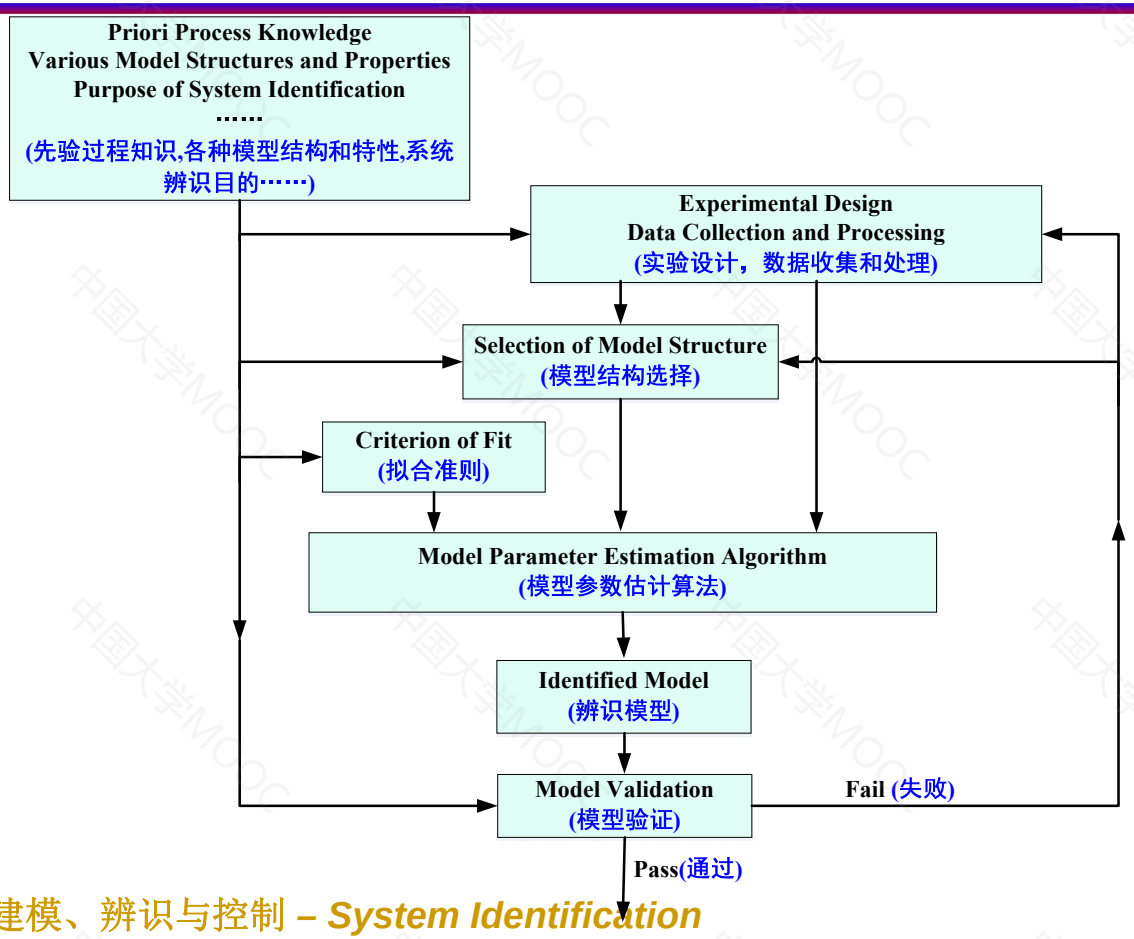
ID Step 4 — Model Validation

- **Is the Identified Model Any Good?** (辨识所得到的模型都是好的吗)
  - A highly subjective task involving scrutinizing all aspects of the model properties (一项高度主观的任务：仔细检查模型特性的所有方面).
  - A good test is to compare the predicted output of the model to measured data that was not used for the fit (called validation data) to study the model's ability to reproduce the behavior of the validation data set by simulation and prediction (好的测试方法是比较模型的预测输出和不用于辨识拟合的测量数据(叫做验证数据)，研究仿真和预测时模型复现验证数据行为的能力).
  - It is also valuable to look at the data that could not be reproduced by the model (the residuals) (可查看不能被模型复现的数据(残差)).
    - Statistically, the residual should not correlate with the system input, indicating the identified model has fully captured the response caused by the system input (统计学上，残差不应该与系统输入相关，表明被辨识模型已经完全捕捉到了系统输入引起的响应)

Requirements of a Good System ID

- Normally undergoes several iterations since various steps are intertwined and some of them are quite subjective (通常需要数次迭代，由于各种步骤交织在一起且一些步骤是非常主观的)
- Practically still heavily depends on the user's understanding of the subject due to the subjectivity involved in experimental design including model structure and identification criterion selection, processing and selecting date sets for parameter estimation (由于实验设计中的主观性而严重依赖于用户对主题的理解，包括模型结构和辨识准则选择、参数估计数据集处理和选择)
  - Knowing the properties of various models and estimation algorithms is the key for a user to make full use of prior information and the observed data in coming out intelligent choices of experimental design, suitable model structure, specific estimation algorithm, and data processing (了解各种模型的特性和估计算法对用户来说是非常重要的，这样他能够完全利用先验信息和观察到的数据以便智能化选择出实验设计、合适的模型结构、特定的估计算法和数据处理方法)

Overall System ID Procedure



Summary

- **Objective of System Identification** (系统辨识的目的)
  - Building mathematical models of dynamical systems based on observed responses of the system (基于观测到的系统响应建立动态系统的数学模型)
- **General Procedure and Common Issues** (一般过程和常见问题)
  - Model structure selection (模型结构选择)
  - Criterion of fit or specific estimation method (拟合准则或确定估计方法）
  - Experimental design and data processing (实验设计和数据处理)
  - Model validation (模型验证)